

PRAKTIKUM SISTEM BASIS DATA

Nama	Ryan Adidaru	No. Modul	6
NPM	2306266994	Tipe	Case Study

1. Implementasi Regex Password & Email:

```
//Regex email dan password di register
const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
if (!emailRegex.test(email)) {
  return res.status(400).json({
    success: false,
    message: "Invalid email format",
    payload: null
  });
}
const passRegex = /^(?=.*\d)(?=.*[a-zA-Z]).{8,}$/;
if (!passRegex.test(password)) {
  return res.status(400).json({
    success: false,
    message: "Invalid password format",
    payload: null
  });
}
```

2. Screenshot bcrypt pada register dan update:

```
}
const hashedPassword = await bcrypt.hash(password, 10);

const existingUser = await userRepository.findByEmail(email);
if (existingUser) {
  return res.status(400).json({
    success: false,
    message: "Email already used",
    payload: null
  });
}
const newUser = await userRepository.create(email, hashedPassword, name);
res.status(201).json({
  success: true,
  message: "User registered",
  payload: newUser
});
catch (error) {
```

```
const hashedPassword = await bcrypt.hash(password, 10);
const updatedUser = await userRepository.update(id, email, hashedPassword, name);

if (updatedUser) {
  res.json({
    success: true,
    message: "User updated",
    payload: updatedUser
  });
} else {
  res.status(404).json({
    success: false,
    message: "User not found",
    payload: null
  });
}
catch (error) {
```

id	name	email	password	balance	created_at
4320f273-01fd-4de0-b72e-1353a36efdc	TestUser	testuser@gmail.com	\$2b\$10\$3cBc0zgJIKjAhh4miALf.uNAX77sGX1vXzB5oXWytq/66HivT1ca	0	2025-03-19 05:08:45.936671

(1 row)

3. Compare hash:

```

    success: false,
    message: "Invalid email or password",
    payload: null
  });
}
const isValidPassword = await bcrypt.compare(password, user.password);
if (!isValidPassword) {
  return res.status(400).json({
    success: false,
    message: "Invalid email or password",
    payload: null
  });
}
res.json({
  success: true,
  message: "Login success",
  payload: user
});
}
```

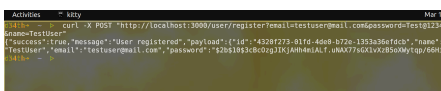
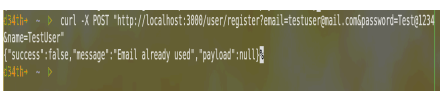
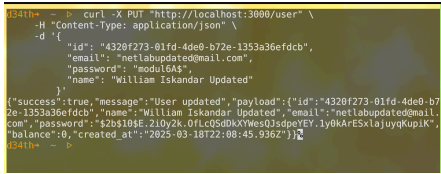
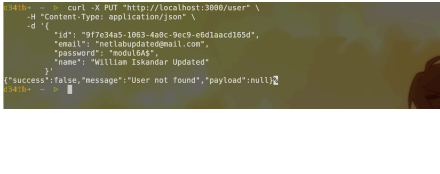
4. CORS Middleware:

```
src > utils > JS corsMiddleware.js > allowedHeaders
1  const express = require('express');
2  const cors = require('cors');
3  const app = express();
4
5  app.use(cors({
6    origin: 'https://os.netlabdte.com',
7    methods: ['GET', 'POST', 'PUT', 'DELETE'],
8    allowedHeaders: ['Content-Type', 'Authorization']
9  }));
10
11 module.exports = app;
12
```

5. Bukti penjalanan query:

```
express_ryan=> CREATE type transaction_status AS ENUM ('pending', 'paid');
CREATE TABLE IF NOT EXISTS transactions(
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
user_id UUID NOT NULL,
item_id UUID NOT NULL,
quantity INT NOT NULL,
total INT NOT NULL,
status transaction_status DEFAULT 'pending',
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
FOREIGN KEY (item_id) REFERENCES items(id) ON DELETE CASCADE
);
CREATE TYPE
CREATE TABLE
express_ryan=> \dt
List of relations
Schema | Name | Type | Owner
public | items | table | neondb_owner
public | stores | table | neondb_owner
public | transactions | table | neondb_owner
public | users | table | neondb_owner
(4 rows)
express_ryan=> █
```

6.

Method, Endpoint	Success	Failed
POST /user/register		
PUT /user		



POST /user/login	<pre>localhost:3000/user/login?email=netlabupdated@gmail.com&password=modul6a \$4f\$ -> curl -X POST "http://localhost:3000/user/login?email=netlabupdated@gmail.com&password=modul6a" {"success":true,"message":"Login success","payload":{"id":"4320f273-01fd-4de0-b72e-1353a36efdcdb","name":"William Iskandar Updated","email":"netlabupdated@gmail.com","password":"\$2b\$10\$6b746f1ErmE90tLg9K.TTublpv14c7BxWQW00v/a1G66L7B0","balance":8,"created_at":"2025-03-18T22:08:45.336Z"}} \$4f\$ -></pre>	<pre>\$34f\$ -> curl -X POST "http://localhost:3000/user/login?email=netlabemail.com&password=modul6" {"success":false,"message":"Invalid email or password","payload":null} \$34f\$ -></pre>
POST /user/topUp	<pre>localhost:3000/user/topUp?userId=4320f273-01fd-4de0-b72e-1353a36efdcdb&amount=1 \$4f\$ -> curl -X POST "http://localhost:3000/user/topUp?userId=4320f273-01fd-4de0-b72e-1353a36efdcdb&amount=1" {"success":false,"message":"Amount must be larger than 0","payload":null} \$4f\$ -></pre>	<pre>\$4f\$ -> curl -X POST "http://localhost:3000/user/topUp?userId=4320f273-01fd-4de0-b72e-1353a36efdcdb&amount=1" {"success":false,"message":"Amount must be larger than 0","payload":null} \$4f\$ -></pre>
POST /transaction/create	<pre>localhost:3000/transaction/create \$4f\$ -> curl -X POST "http://localhost:3000/transaction/create" {"success":true,"message":"Transaction created","payload":{"id":"1a3f4867-bb0a-4d2a-be6d-8ec5b7effc83","user_id":"4320f273-01fd-4de0-b72e-1353a36efdcdb","item_id":"123e4567-e89b-12d3-a456-426614174000","quantity":1,"total":15000,"status":"pending","created_at":"2025-03-18T23:39:29.837Z"}} \$4f\$ -></pre>	<pre>\$39:29.837Z"} \$4f\$ -> curl -X POST "http://localhost:3000/transaction/create" {"success":false,"message":"Quantity must be larger than 0","payload":null} \$4f\$ -></pre>
POST /transaction/pay	<pre>localhost:3000/transaction/pay?transactionId=1a3f4867-bb0a-4d2a-be6d-8ec5b7effc83 \$4f\$ -> curl -X POST "http://localhost:3000/transaction/pay/1a3f4867-bb0a-4d2a-be6d-8ec5b7effc83" {"success":true,"message":"Payment successful","payload":{"id":"1a3f4867-bb0a-4d2a-be6d-8ec5b7effc83","user_id":"4320f273-01fd-4de0-b72e-1353a36efdcdb","item_id":"123e4567-e89b-12d3-a456-426614174000","quantity":1,"total":15000,"status":"paid","created_at":"2025-03-18T23:39:29.837Z"}} \$4f\$ -></pre>	<pre>localhost:3000/transaction/pay?transactionId=1a3f4867-bb0a-4d2a-be6d-8ec5b7effc83 \$4f\$ -> curl -X POST "http://localhost:3000/transaction/pay/1a3f4867-bb0a-4d2a-be6d-8ec5b7effc83" {"success":false,"message":"Insufficient balance","payload":null} \$4f\$ -></pre>
DELETE /transaction/:id	<pre>localhost:3000/transaction/:id \$4f\$ -> curl -X DELETE "http://localhost:3000/transaction/1a3f4867-bb0a-4d2a-be6d-8ec5b7effc83" {"success":true,"message":"Transaction deleted successfully","payload":{"id":"1a3f4867-bb0a-4d2a-be6d-8ec5b7effc83","user_id":"4320f273-01fd-4de0-b72e-1353a36efdcdb","item_id":"123e4567-e89b-12d3-a456-426614174000","quantity":1,"total":15000,"status":"paid","created_at":"2025-03-18T23:39:29.837Z"}} \$4f\$ -></pre>	<pre>localhost:3000/transaction/:id \$4f\$ -> curl -X DELETE "http://localhost:3000/transaction/1a3f4867-bb0a-4d2a-be6d-8ec5b7effc83" {"success":false,"message":"Transaction not found","payload":null} \$4f\$ -></pre>